

JPE Replication Report

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1 Paper Identification

ID	20241468
Author	Medici
Title	Closing Ranks: Organized Labor and Immigration
Iteration	2

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. We have a few missing data citations to add. See below.
2. The confidential part run on your machine was a success, all good.
3. There are quite a few discrepancies. Please check each of them and advise what is wrong. Hard to say what could be the reason for those, please check.

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☐ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☐ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☐ Dataset list

- ☒ Computational requirements
 - ☒ Software Requirements
 - ☐ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☒ License for Code (Optional, but recommended)
- ☒ {REPLICATOR} to Replicators
 - ☒ Details (as necessary)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☐ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. – As per the [JPE policy](#), "In cases where data cannot be published in an openly accessible trusted data repository like the

JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication."

{REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

- Table E.1 is built with Crouch (1993) data. Missing data source in the README.md
- Same for Table F.1 built with Austrian National Library (2024), Mackie and Rose (2016), and Nohlen and Stöver (2010).

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
IPUMS USA complete count (1880–1930)	yes	no	yes	yes	yes	yes
Haines (2010), ICPSR 02896	yes	no	yes	no	yes	yes
ICPSR 00001 (elections)	no	no	yes	no	no	no
IPUMS restricted full-count with surnames	no	yes	yes	yes	yes	yes
Union proceedings	yes	no	no	no	no	no
Gregory (2015), IWW History Project	yes	no	yes	no	yes	yes
Garlock (2009) / KoL, ICPSR 00029	no	yes	yes	no	yes	yes
Fed. Reserve Bank of	yes	no	yes	no	yes	no

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
Minneapolis, CPI						
Migration Policy Institute	yes	no	yes	no	yes	no
Willcox (1929)	yes	no	no	no	yes	yes
Freeman (1998), NBER WP 6012	yes	no	yes	no	yes	yes
Ferrara et al. (2022), ICPSR 150101	yes	no	yes	no	yes	yes
IPUMS NHGIS, 1930 county shapefile	yes	no	no	no	yes	yes
Day (1892), mineral industries	yes	no	no	no	yes	yes
Atack (2016), railroad GIS	yes	no	yes	yes	yes	yes
Farber et al (2021), Weather_Data.dta	yes	no	no	no	yes	yes
Sequeira et al (2020), unionshares.dta	yes	no	no	no	yes	yes

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☒ Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

6.1 Stated Requirements

- ☐ No requirements specified

☒ Software Requirements specified as follows:

- Stata 19.5
- Required packages:
 - `reghdfe` 6.13.1
 - `ftools` 2.50.0
 - `gtools` 1.11.8
 - `ivreghdfe` 1.1.3
 - `ivreg2` 4.1.12
 - `ranktest` 2.0.04
 - `avar` 1.0.07
 - `weakivtest` no stamp
 - `weakivtest2` no stamp
 - `acreg` 1.1.0
 - `hdfe` 3.2.9
 - `distinct` 1.2.1
 - `winsor2` 1.1
 - `binscatter` 7.02
 - `outreg2` 2.3.2
 - `estout` 3.33
 - `coefplot` 1.8.6
 - `grc1leg2` 2.26
 - `grstyle` 1.1.1
 - `fre` 1.2.5
 - `spmap` 1.3.2
 - `shp2dta` 1.0.9
 - `geoinpoly` 2.0.2
 - `palettes`
 - `colrspace`
 - `egenmore`

☒ Computational Requirements specified as follows:

- Windows Server 2022 Standard (Version 21H2), 64-bit operating system, Intel(R) Xeon(R) Gold 6426Y 2.50 GHz (2 processors), 1TB RAM.
- Disk space: about 10 GB free

☒ Time Requirements specified as follows:

- 12.5 hours

☐ Requirements are complete.

For missing requirements, see the list of required changes in [Section 9](#)

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary:

- Data files with PII indicators: 21
- Variables flagged in data: 114
- Code files with PII references: 39
- PII references in code: 2663

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	Crosswalk_1880_1930.dta	1	lat
Data	Crosswalk_1890_1930.dta	1	lat
Data	Crosswalk_1900_1930.dta	1	lat
Data	Crosswalk_1910_1930.dta	1	lat
Data	Crosswalk_1920_1930.dta	1	lat
Data	IWW_unions_county1930.csv	1	loc
Data	KoL_unions_county1930.csv	1	loc
Data	Railroad_Ataack_1930countyboundaries.csv	1	name
Data	Weather_Data.dta	8	son, lat, lon, country, name
Data	agri_census_1890.csv	1	name
Data	coalmines_1890.csv	2	name, loc
Data	electionresults_1856_county.csv	1	name
Data	electionresults_1886-1924_county.csv	1	name
Data	mfg_census.csv	1	name
Data	mines_census_1890.csv	2	name, loc

File Type	File	Variables/References	PII Categories
Data	names_ancestry.dta	26	name
Data	names_origin.dta	27	name
Data	unions_combined_county1930.csv	34	loc
Data	willcox_immigration_bycountry.csv	1	country
Data	xwalk_occ1950_occnames.csv	1	name

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-07-10T13:45:51.550

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any windows filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that STATA allows / to be a filepath separator on a windows platform, hence this is a requirement for all STATA applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (C:\ etc)
39	0	6	0	0

7.1.3 Duplicate Files Report

We did not find any duplicate files.

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
/MS20241468_Deposit/data/public/Sequeira_et_al_2020/Weather_Data.dta	218.78
/MS20241468_Deposit/data/public/IPUMS_surname_aggregates/names_ancestry.dta	1326.55
/MS20241468_Deposit/data/public/IPUMS_surname_aggregates/names_origin.dta	3198.84

7.2 Code description

There are 16 `.do` files, one of which is the master.

cloc		github.com/AIDanial/cloc v 2.06 T=0.59 s (262.9 files/s, 520575.1 lines/s)		
Language	files	blank	comment	code
CSV	57	0	0	290744
Stata	37	2610	52	10209
Markdown	21	210	0	791
Text	2	35	0	256
XML	36	0	0	36
TeX	1	0	0	26
---	---	---	---	---
SUM:	154	2855	52	302062

☒ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

7.3 Computing Environment of the Replicator

- Nuvolos: Ubuntu 24.04.1 LTS, 16 vCPU + 64 GB RAM
- Stata/MP 19.5

8 Replication

- Downloaded code and data from dropbox.
- Copied git repo and unzipped the replication package into this repo.
- Adjusted the root of the `00_Master.do` as per `README.md`

- Ran code as per README. Code broke because of missing packages `center`. Ran again.
- Ran code as per README. Code broke because of missing packages `require`. Ran again.
- Ran code as per README. Code broke because of missing packages `ivreg_ss`. Ran again.
- Code ran as expected. Differences in output. See [Section 9](#)
- Upon agreement with the author, run the do-files for the part with confidential data contained in `build_documentation/scripts`.
- After adjusting the root path in the do-file `MS20241468_Deposit/build_documentation/scripts/00_master_build.do` and `MS20241468_Deposit/build_documentation/scripts/09_master_build.do`, code broke with the error:

47. file U:/Downloads/Carlo Medici -

MS20241468_JPE_Replication_20260708/MS20241468_Deposit/data/_raw_local/IPUMS/IPUMS_fullcount_1880.dta not found
r(601);

- Adjusted the root path in the do-file `MS20241468_Deposit/build_documentation/scripts/build/b9_names.do` and code run without errors.
- We interrupted the session after 60 minutes and continued the day after. In the continuing session, we checked the PID of the Stata window and uploaded the folders `MS20241468_Deposit/data/intermediate` and `MS20241468_Deposit/data/public` as specified by the `MS20241468_Deposit/build_documentation/README.md` to be the only one affected by those scripts.
- After running the `b**` scripts, checked all the files generated using `KDiff3`. All dataset matched. No difference was detected.

9 Findings

9.1 Missing Requirements

- ☒ Software Requirements
 - ☒ Stata
 - ☒ Missing required packages: `center`, `require`, `ivreg_ss`
- ☐ Computational Requirements specified as follows:
 - ☐ Missing needed hard-disk memory
- ☐ Time Requirements
 - Length of necessary computation (hours, weeks, etc.)

[REQUIRED] Please amend README to contain complete requirements.

9.2 Data Preparation Code

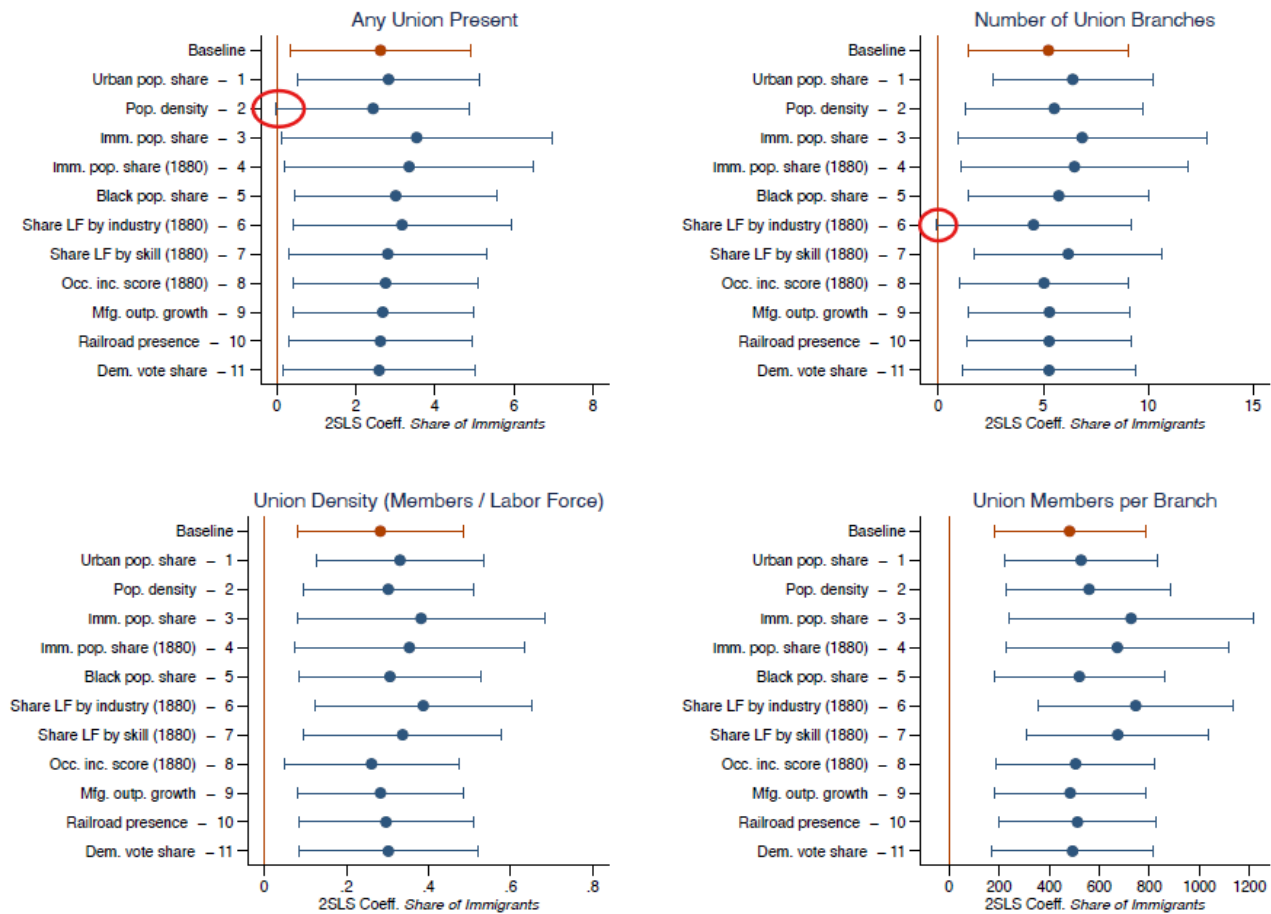
9.3 Tables and Figures

Figure/Table #	Program	Line Number	Reproduced?
Figure 1	2b_figures_descriptives.do	66	yes
Figure 2	2a_sumstats.do	50	yes
Figure 3	2e_results_robustness.do	1536	yes
Figure 4	2e_results_robustness.do	229	no, see Figure 1
Figure 5	2a_sumstats.do	427	yes
Table 1	2a_sumstats.do	155	no, see Figure 7
Table 2	2c_results_main.do	42	no, see Figure 8
Table 3	2c_results_main.do	80	no, see Figure 9
Table 4	2c_results_main.do	205	no, see Figure 10
Table 5	2c_results_main.do	289	no, see Figure 11
Table 6	2c_results_main.do	342	no, see Figure 12
Table 7	2c_results_main.do	484	no, see Figure 13
Table 8	2c_results_main.do	560	no, see Figure 14
Figure A.1	2b_figures_descriptives.do	40	yes
Figure A.2	/		
Figure A.3	/		
Figure A.4	2a_sumstats.do	285	no, see Figure 2
Figure A.5	2c_results_main.do	808	no, see Figure 3
Figure B.1	1e_weather_shocks.do	189	no, see Figure 4

Figure/Table #	Program	Line Number	Reproduced?
Figure B.2	2e_results_robustness.do	221	no, see Figure 5
Figure G.1	2f_rottemberg_weights.do	254	no, see Figure 6
Figure H.1	2a_sumstats.do	355	yes
Table A.1	/		
Table A.2	2a_sumstats.do	199	yes
Table A.3	2a_sumstats.do	236	no, see Figure 15
Table A.4	2d_results_extra.do	38	no, see Figure 16
Table A.5	2c_results_main.do	409	no, see Figure 17
Table A.6	2c_results_main.do	759	no, see Figure 18
Table A.7	2c_results_main.do	858	no, see Figure 19
Table A.8	2d_results_extra.do	94	no, see Figure 20
Table A.9	2d_results_extra.do	149	no, see Figure 21
Table A.10	2d_results_extra.do	210	no, see Figure 22
Table A.11	2d_results_extra.do	278	no, see Figure 23
Table B.1	2e_results_robustness.do	1137	no, see Figure 24
Table B.2	2e_results_robustness.do	44	yes
Table E.1	/		no, missing
Table F.1	/		no, missing
Table G.1	2f_rottemberg_weights.do	196	no, see Figure 25

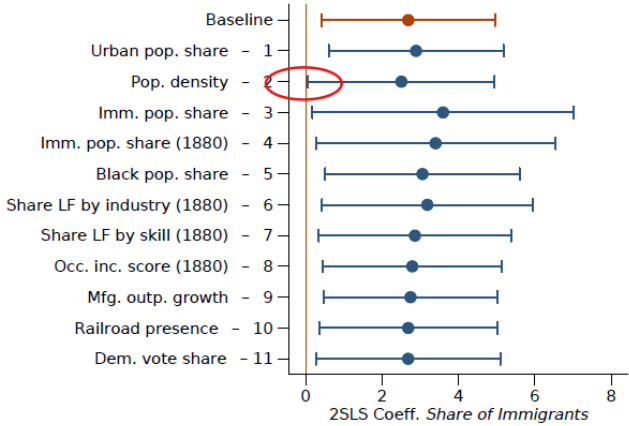
Figure/Table #	Program	Line Number	Reproduced?
Table G.2	2e_results_robustness.do	44	yes

- Figure 4: differences in confidence intervals

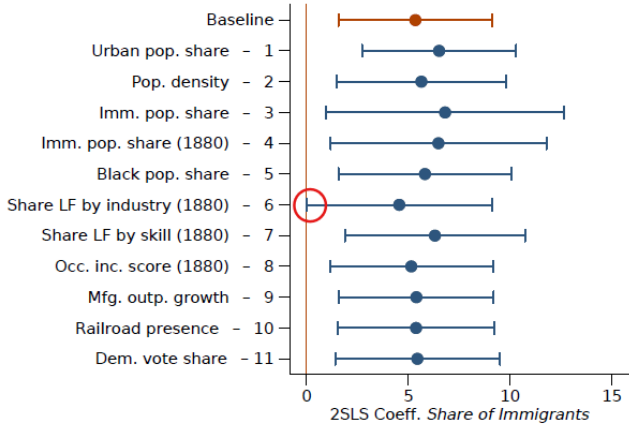


Paper version

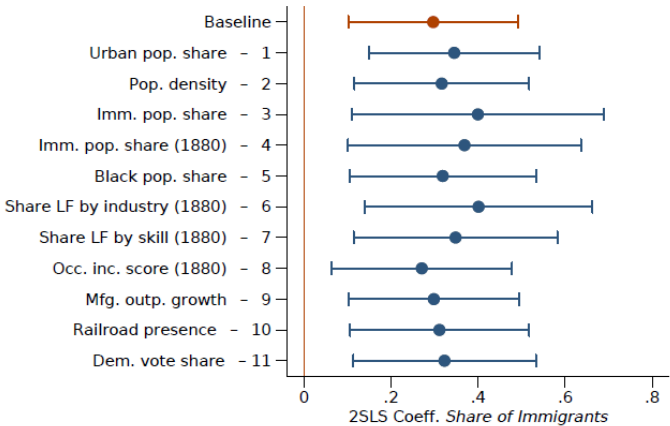
Any Union Present



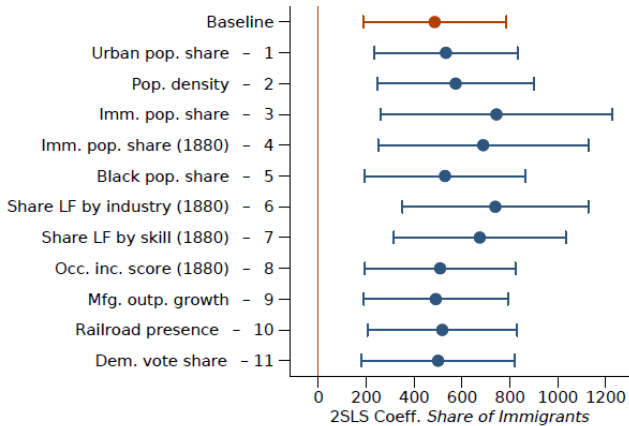
Number of Union Branches



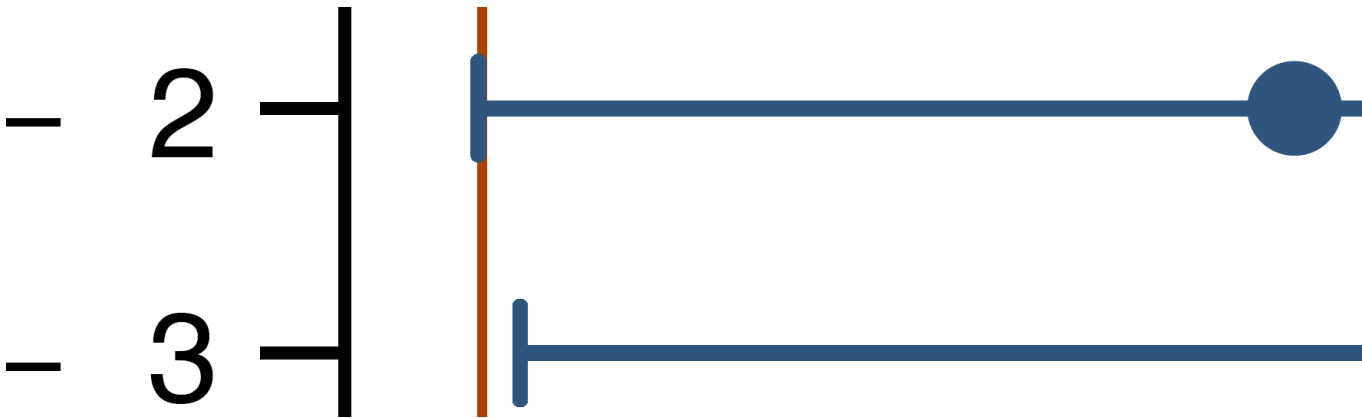
Union Density (Members / Labor Force)



Union Members per Branch



Replicated version



Paper version, detail 1

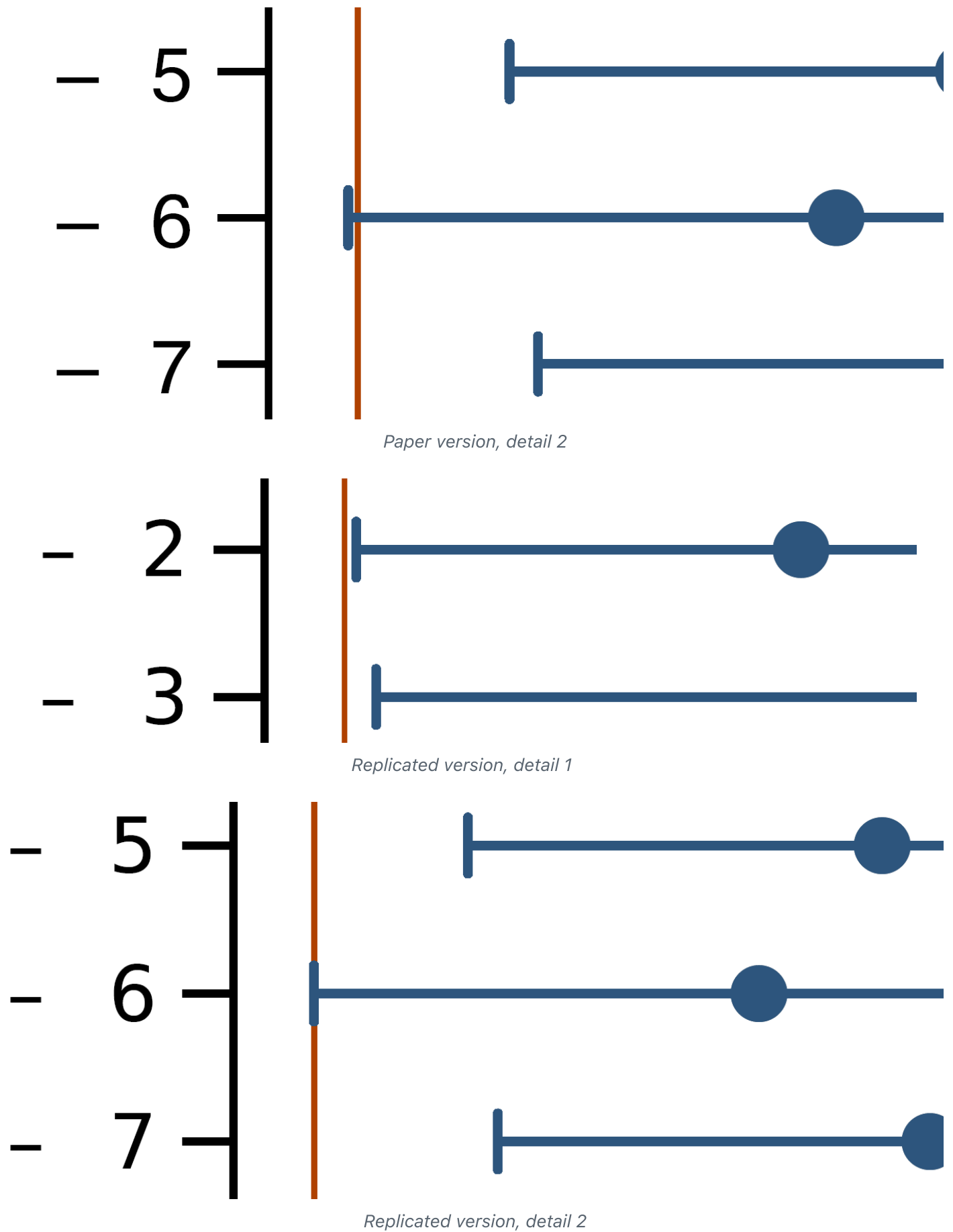
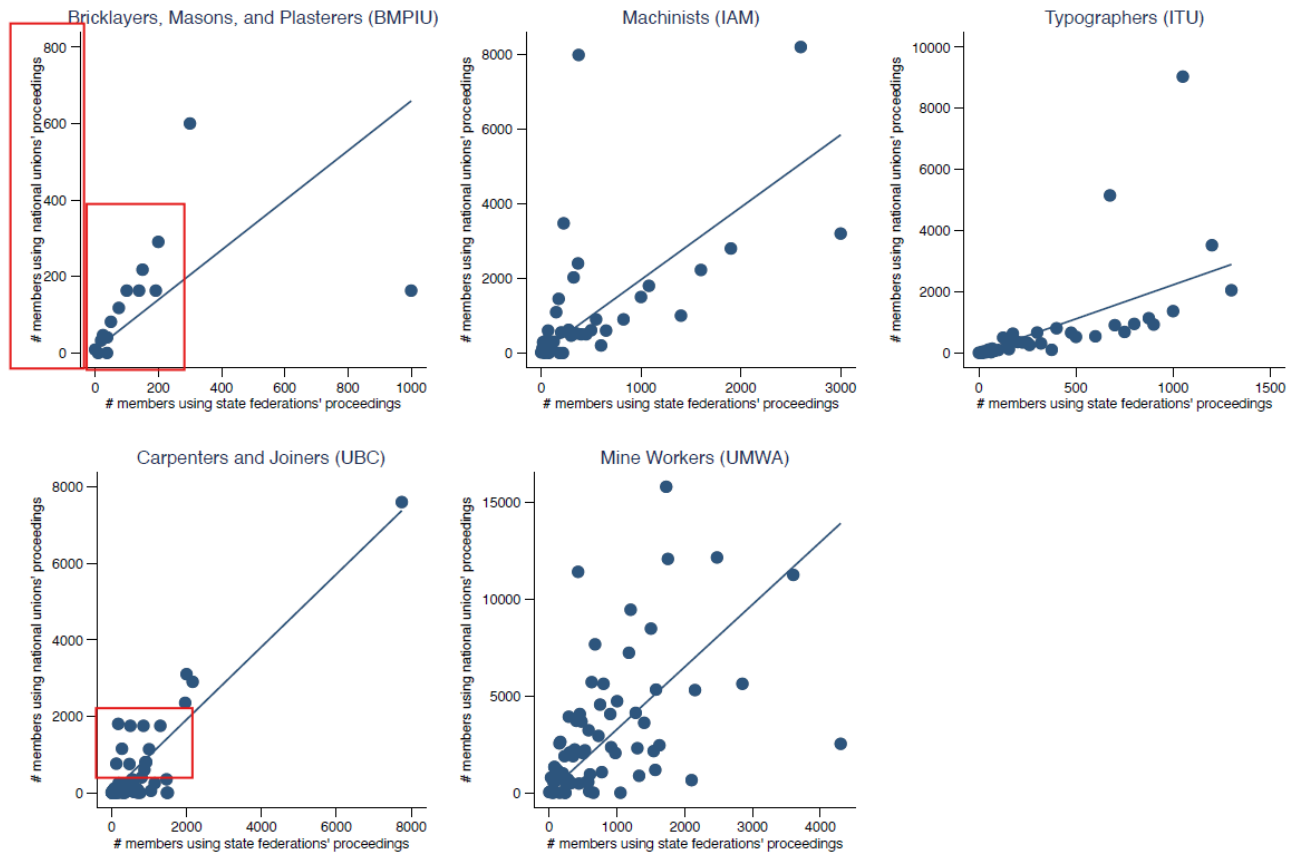


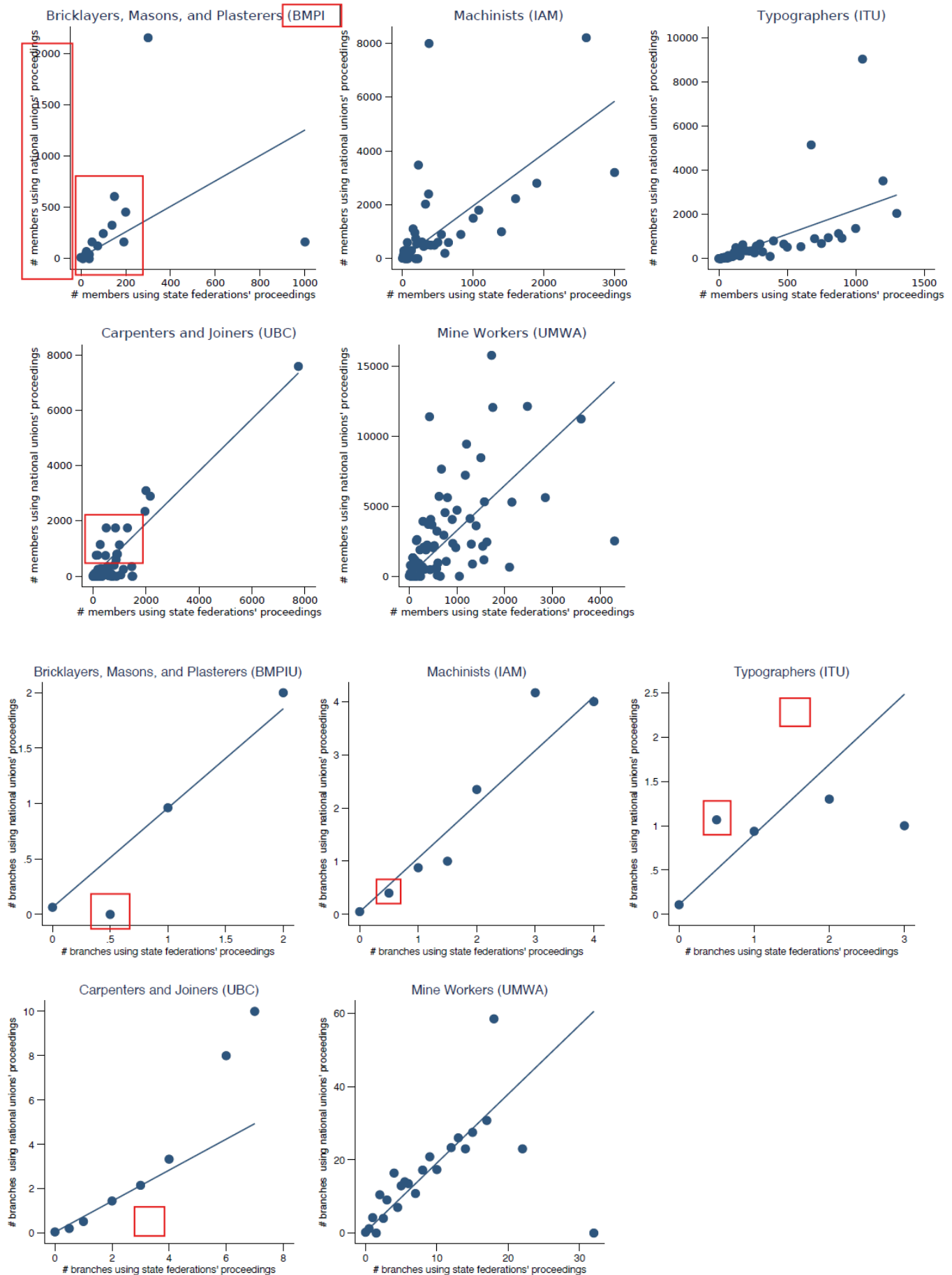
Figure 1: Figure 4

- Figure A4: differences in some panels



Paper version

Panel A: Number of Union Members



Panel B: Number of Union Branches

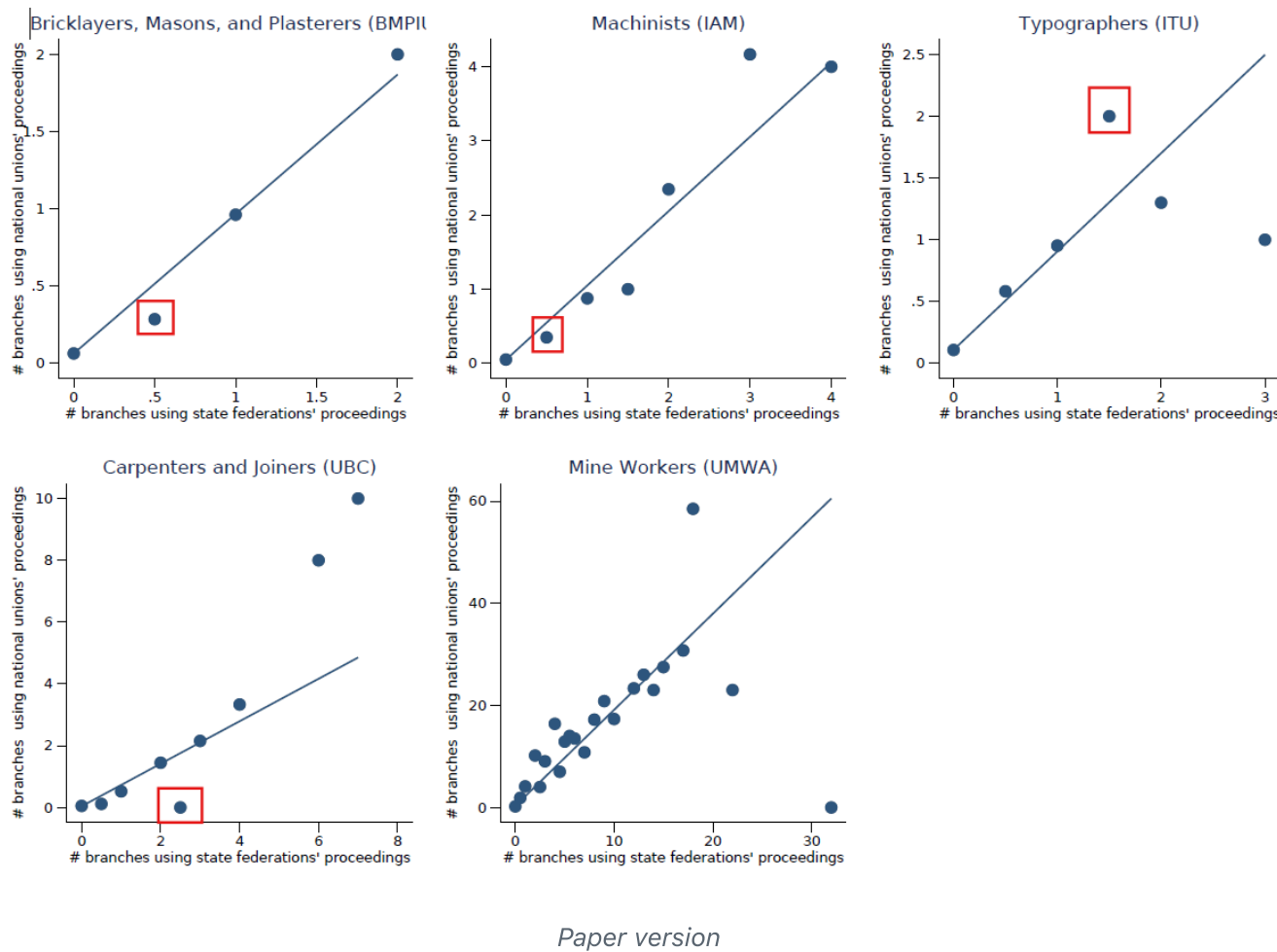


Figure 2: Figure A4

- Figure A5: differences in scaling of the right panel

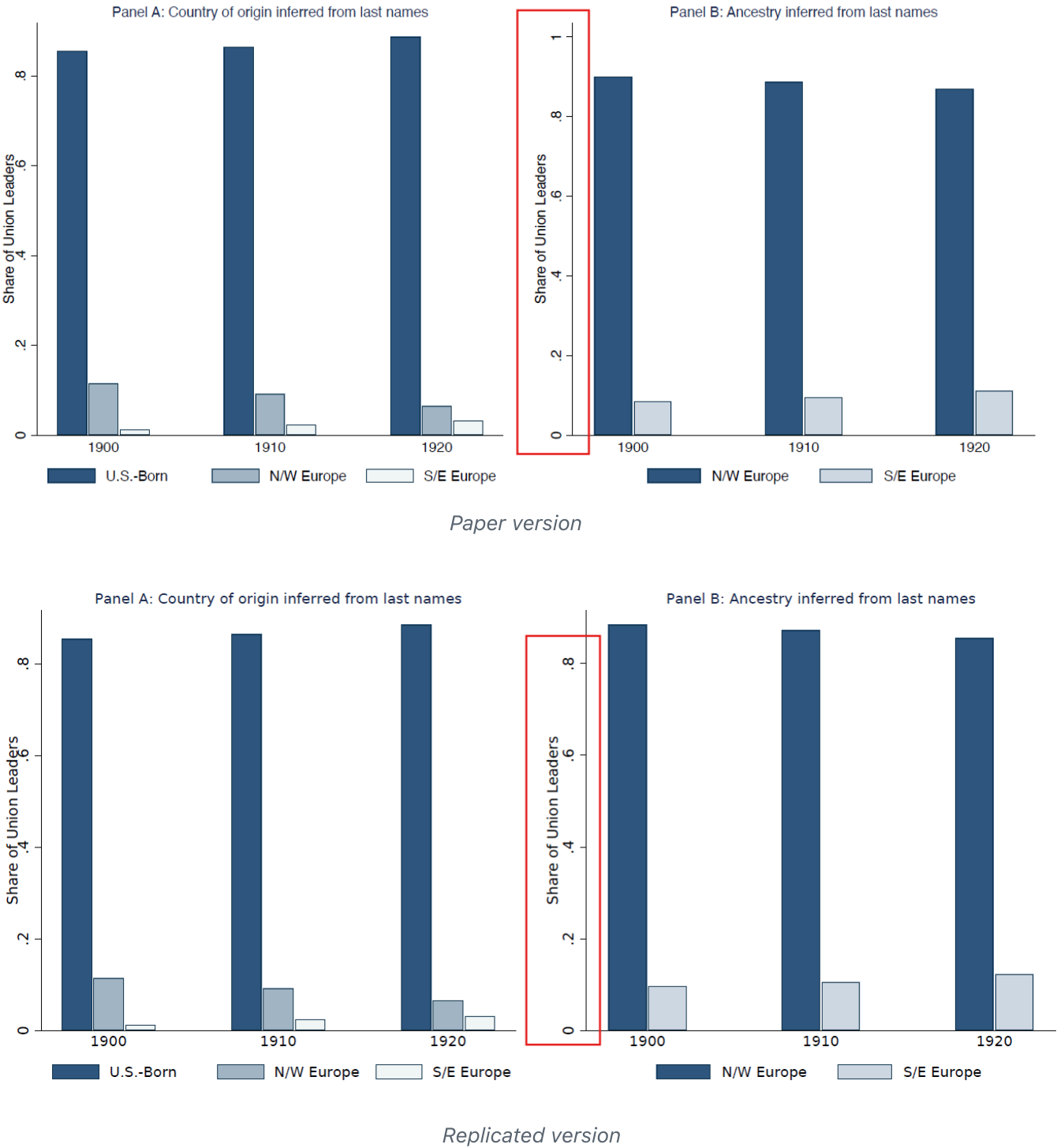
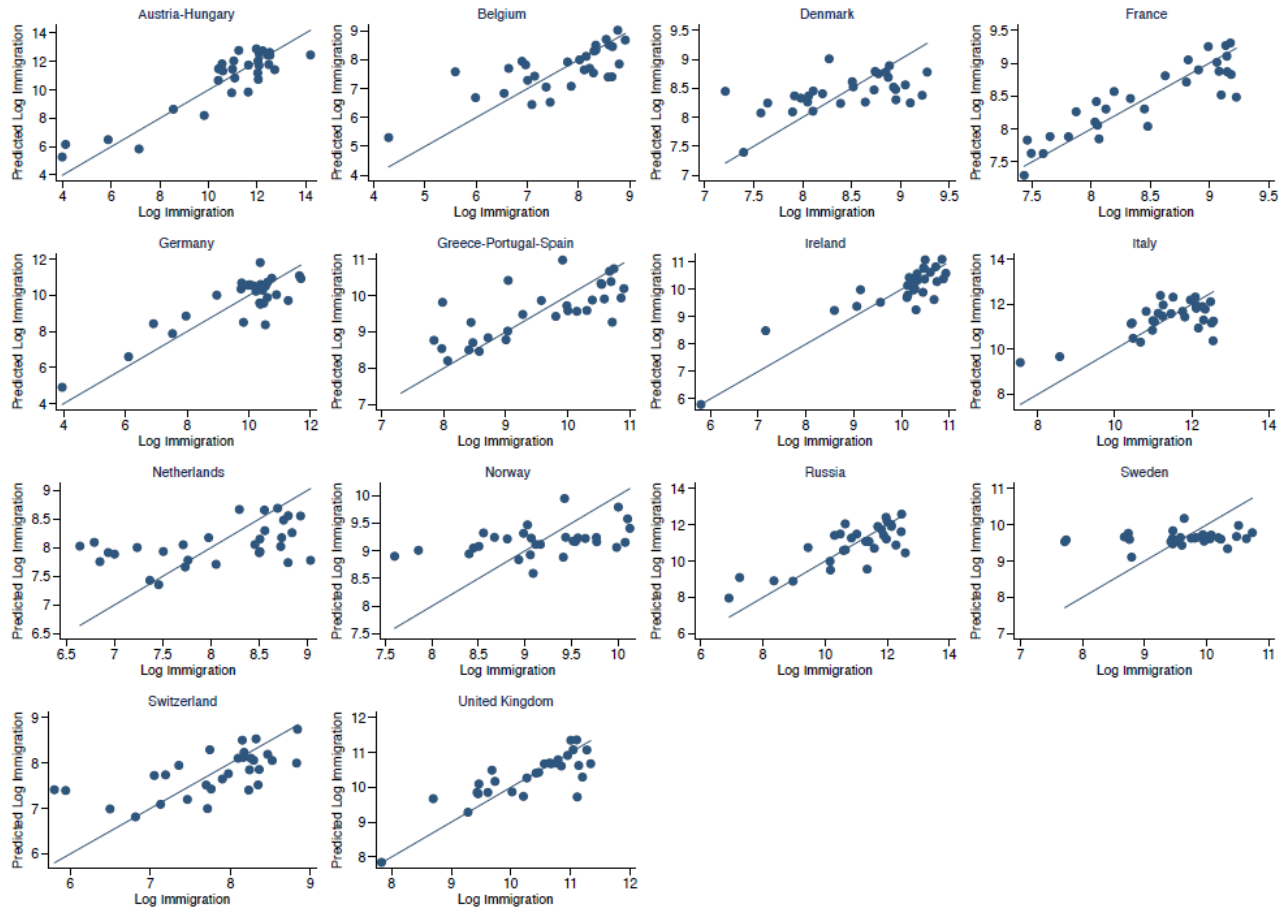
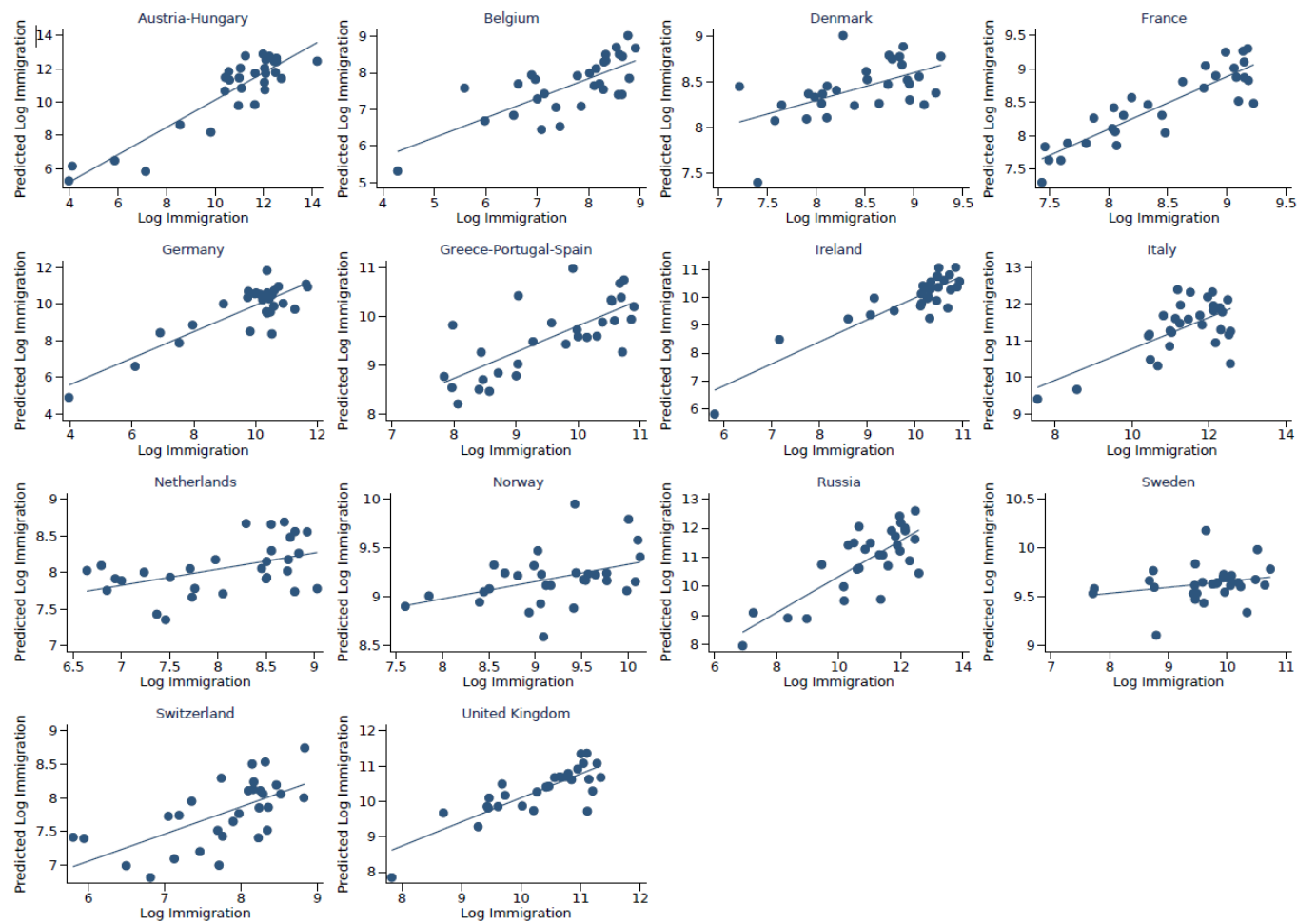


Figure 3: Figure A5

- Figure B1: differences in some panels



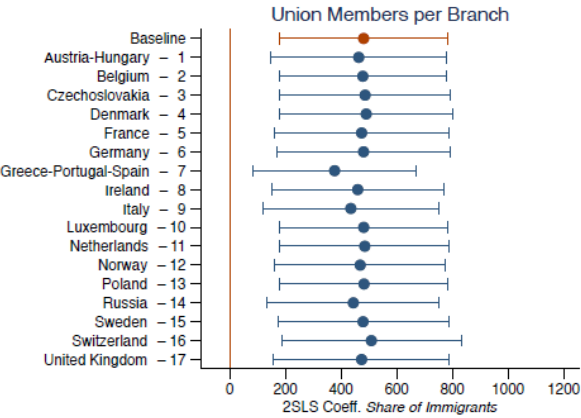
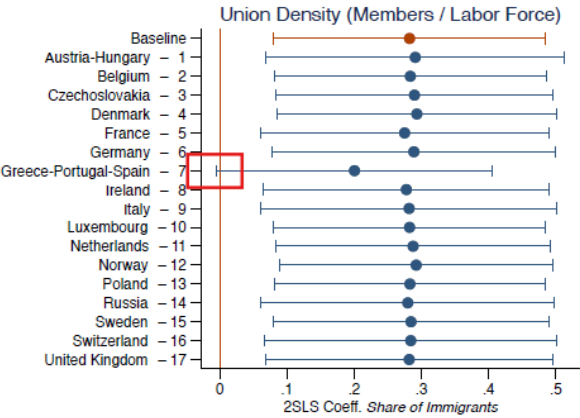
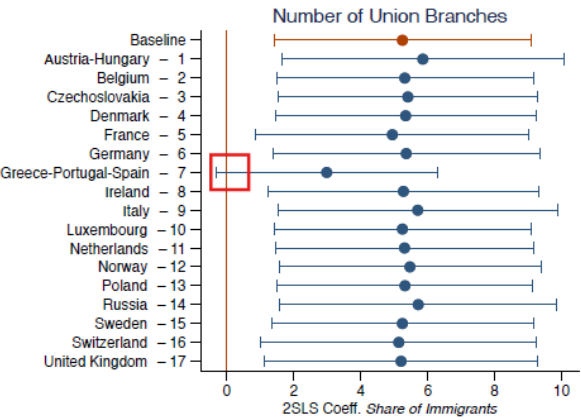
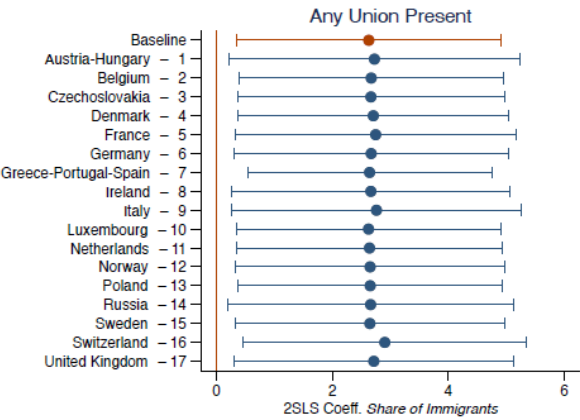
Paper version



Replicated version

Figure 4: Figure B1

- Figure B2: differences in confidence intervals



Paper version

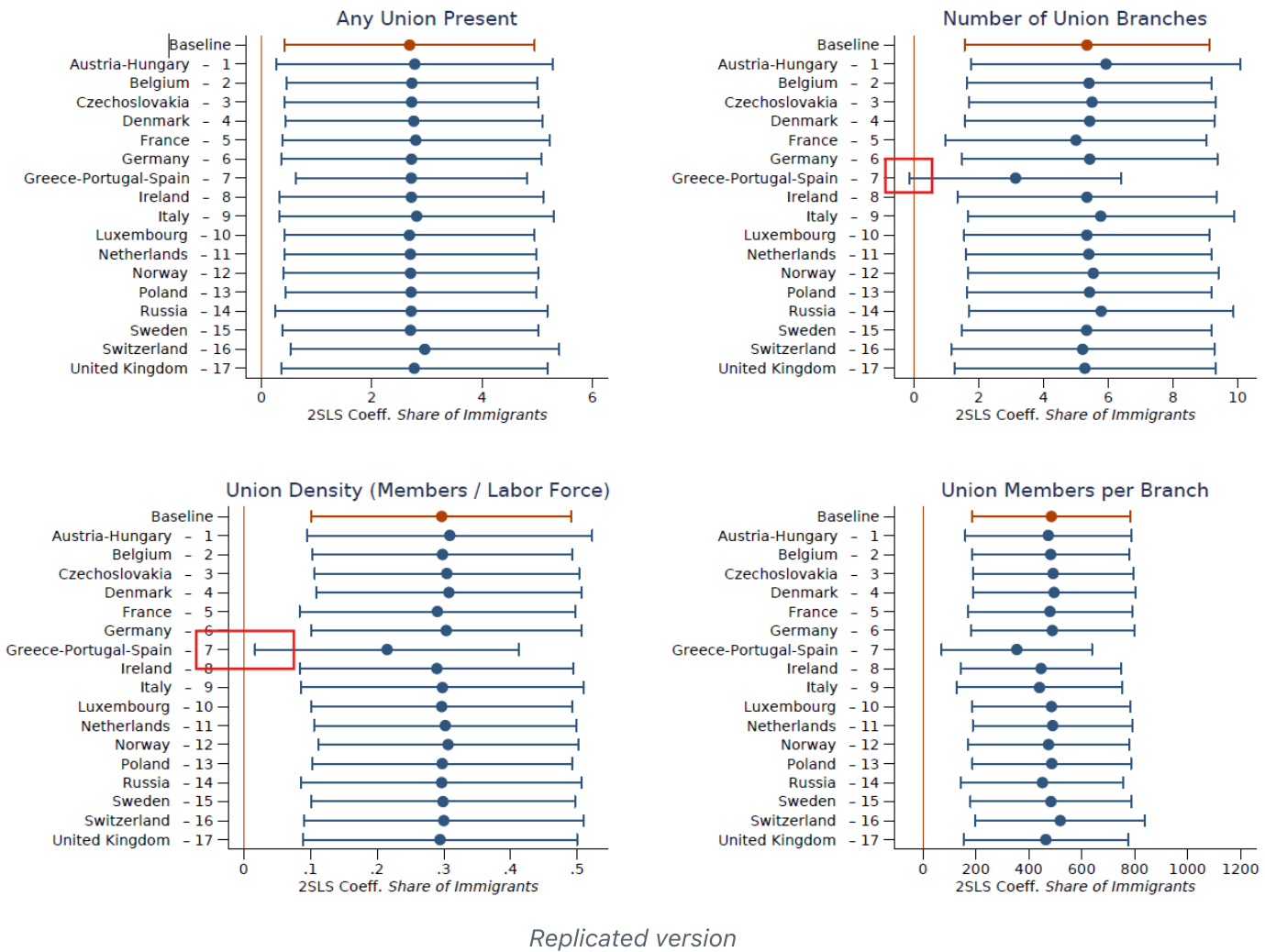
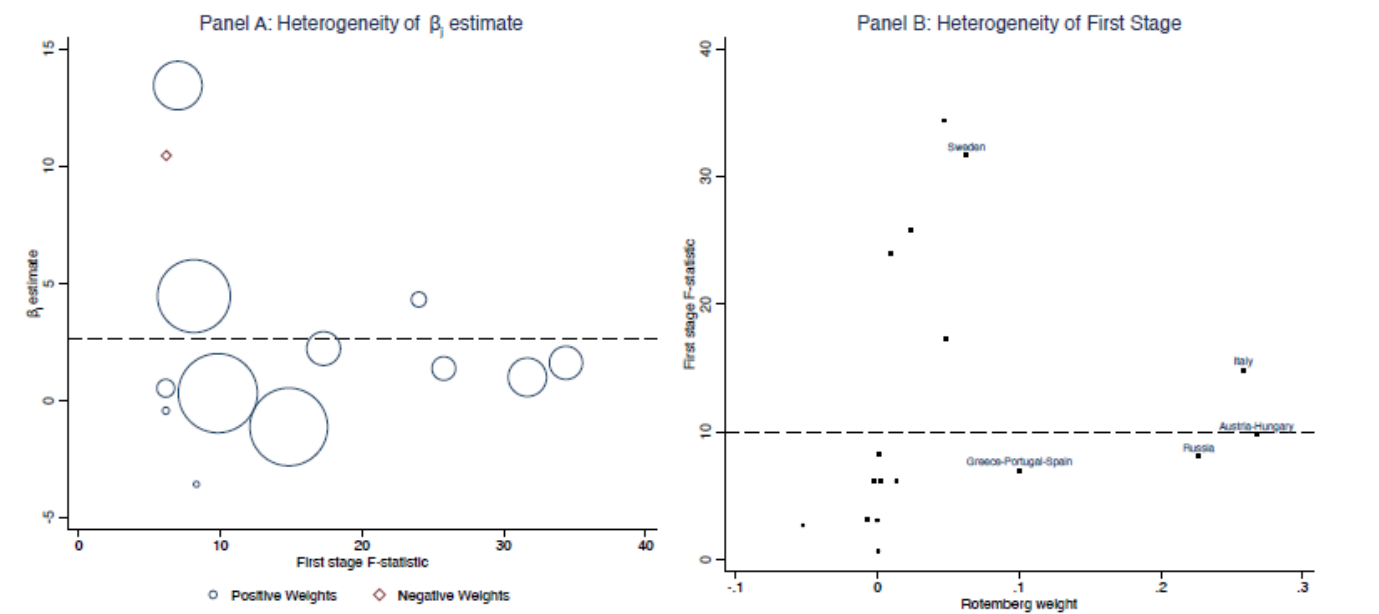
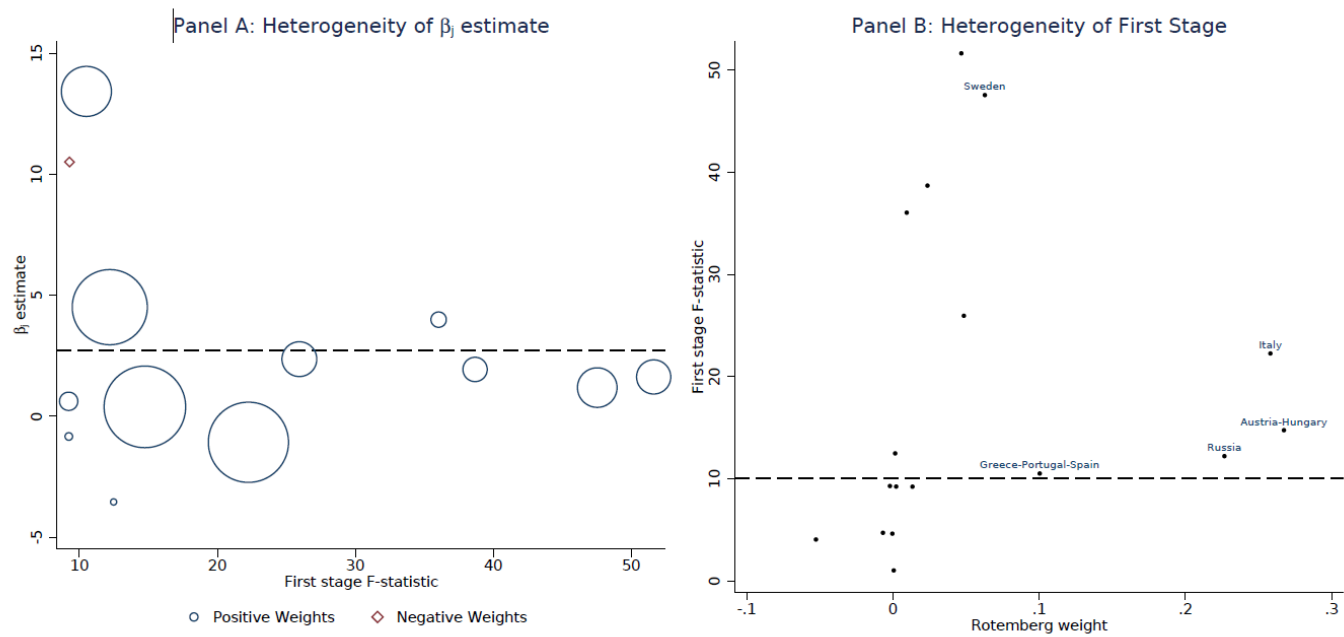


Figure 5: Figure B2

- Figure G1: different replicated output



Paper version



Replicated version

Figure 6: Figure G1

- Table 1: different replicated output

	Obs.	Mean	St. Dev.
<i>Panel A: Unionization</i>			
Any Union Present	2,628	0.44	0.50
Number of Union Branches	2,628	2.56	5.30
Union Density (Members / Labor Force)	2,628	0.04	0.08
Union Members per Branch	2,628	48.16	74.38
<i>Panel B: Demographics</i>			
Share of Urban Population	2,628	0.30	0.27
Share of Immigrant Population	2,628	0.09	0.10
Share of European Immigrant Population	2,628	0.08	0.08
Share of European Imm. Pop. (<10 years in U.S.)	2,628	0.02	0.03
<i>Panel C: Labor Force (men ages 16–64)</i>			
Total Labor Force	2,628	14,777.32	44,231.97
Labor Force Participation Rate	2,628	0.91	0.04
Share of U.S.-Born Labor Force	2,628	0.86	0.15
Share of European Immigrant Labor Force	2,628	0.12	0.12
<i>Panel D: Baseline Characteristics (in 1890)</i>			
Share of European Immigrant Population	876	0.10	0.10
Share of Population in Manufacturing	876	0.03	0.04
Share of Population in Agriculture	876	0.45	0.19
Presence of Active Coal Mines	876	0.26	0.44

Paper version

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	N	mean	sd	N	mean	sd
afl_comb_all_pres				2,628.00	0.44	0.50
afl_comb_all_locals				2,628.00	2.59	5.33
afl_comb_all_dens				2,628.00	0.04	0.08
afl_comb_all_avgmemb				2,628.00	48.58	74.41
all_totpop_mw				2,628.00	49,494.70	138,738.48
urban_share_mw				2,628.00	0.30	0.27
imm_share_mw				2,628.00	0.09	0.10
euro_imm_share_mw				2,628.00	0.08	0.08
sh_euro_10yr_mw				2,628.00	0.02	0.03
all_lf_tot_m				2,628.00	14,777.32	44,231.97
lfpartrate_m				2,628.00	0.91	0.04
share_lf_nat				2,628.00	0.86	0.15
share_lf_euro_imm				2,628.00	0.12	0.12
euro_imm_share_mw_1890	876.00	0.10	0.10			
farmfams_share_1890	876.00	0.45	0.19			
mfglabor_share_mw_1890	876.00	0.03	0.04			
mining_1890	876.00	0.26	0.44			

Replicated version

Figure 7: Table 1

- Table 2: different replicated output

Table 2: First Stage of the Instrumental Variable Estimation

	Dependent variable: Share of Immigrants			
	(1)	(2)	(3)	(4)
Predicted Share of Immigrants	0.336*** (0.037)	0.304*** (0.036)	0.274*** (0.033)	0.273*** (0.033)
Observations	2628	2628	2628	2628
Dep. var. mean	0.028	0.028	0.028	0.028
Indep. var. mean	0.030	0.030	0.030	0.030
KP F-statistic	80.67	71.09	67.76	67.34
Controls:				
Share of Population in Manufacturing	No	Yes	Yes	Yes
Share of Population in Agriculture	No	No	Yes	Yes
Presence of Coal Mines	No	No	No	Yes

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	Immigrant Share	Immigrant Share	Immigrant Share	Immigrant Share
pr1890_sh_euro_wa_10yr_m	0.336*** (0.037)	0.304*** (0.036)	0.274*** (0.033)	0.273*** (0.033)
Nr. observations	2628	2628	2628	2628
Outcome mean	0.028	0.028	0.028	0.028
Imm. Share mean	0.030	0.030	0.030	0.030
KP F-stat	80.70	71.12	67.78	67.36
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	N	Y	Y	Y
Share Agriculture	N	N	Y	Y
Active Coal Mines	N	N	N	Y

Replicated version

Figure 8: Table 2

- Table 3: different replicated output

Table 3: The Effect of Immigration on Organized Labor

	Dependent variable:			
	Any Union Present (1)	Number of Union Branches (2)	Union Density (Members / LF) (3)	Union Members per Branch (4)
<i>Panel A: OLS</i>				
Share of Immigrants	0.630* (0.374)	2.089*** (0.622)	0.090* (0.048)	171.230*** (58.821)
<i>Panel B: Reduced Form</i>				
Predicted Share of Immigrants	0.716** (0.284)	1.430*** (0.488)	0.077*** (0.029)	131.170*** (40.503)
<i>Panel C: 2SLS</i>				
Share of Immigrants	2.626** (1.157)	5.245*** (1.952)	0.282*** (0.103)	481.007*** (153.851)
Observations	2,628	2,628	2,628	2,628
Dep. var. mean	0.441	2.563	0.036	48.158
Indep. var. mean	0.028	0.028	0.028	0.028
KP F-statistic	67.34	67.34	67.34	67.34

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
Immigrant Share	0.621* (0.345)	2.162*** (0.577)	0.099** (0.048)	163.137*** (56.502)
Nr. observations	2628	2628	2628	2628
Dep. var. mean	0.444	2.589	0.037	48.577
Indep. var. mean	0.028	0.028	0.028	0.028
Estimator	OLS	OLS	OLS	OLS
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel A

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
pr1890_sh_euro_wa_10yr_m	0.733*** (0.283)	1.457*** (0.481)	0.081*** (0.028)	132.411*** (40.225)
Nr. observations	2628	2628	2628	2628
Dep. var. mean	0.444	2.589	0.037	48.577
Indep. var. mean	0.028	0.028	0.028	0.028
Estimator	Reduced Form	Reduced Form	Reduced Form	Reduced Form
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel B

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
Immigrant Share	2.687** (1.156)	5.344*** (1.925)	0.297*** (0.099)	485.556*** (152.350)
Nr. observations	2628	2628	2628	2628
Dep. var. mean	0.444	2.589	0.037	48.577
Indep. var. mean	0.028	0.028	0.028	0.028
KP F-stat	67.36	67.36	67.36	67.36
Estimator	2SLS	2SLS	2SLS	2SLS
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel C

Figure 9: Table 3

- Table 4: different replicated output

TABLE 4. EFFECTS ON THE INTENSIVE AND EXTENSIVE MARGINS

	Number of Union Branches (1)	Union Density (Members / LF) (2)	Union Members per Branch (3)
<i>Intensive Margin:</i> Share of Immigrants $\times$ (Already Unions = 1)	4.857** (2.059)	0.227* (0.117)	476.494*** (180.057)
<i>Extensive Margin:</i> Share of Immigrants $\times$ (Already Unions = 0)	5.744** (2.447)	0.353*** (0.122)	486.807*** (168.862)
Observations	2,628	2,628	2,628
KP F-statistic	33.31	33.31	33.31
LM F-statistic	35.95	35.95	35.95
SW F-statistic (Already Unions = 1)	68.05	68.05	68.05
SW F-statistic (Already Unions = 0)	57.98	57.98	57.98

Paper version

	(1)	(2)	(3)
VARIABLES	_comb_all_ihs	_comb_all_deomb	_all_avgmemb
sheuro_intm_all	4.936** (2.009)	0.245** (0.110)	486.213*** (177.638)
sheuro_extm_all	5.867** (2.438)	0.363*** (0.122)	484.713*** (168.560)
Observations	2,628	2,628	2,628
Outcome mean	2.589	0.037	48.577
Share Immigrants mean	0.028	0.028	0.028
Always Unions mean	0.314	0.314	0.314
KP F-stat	33.32	33.32	33.32
LM F-stat	35.94	35.94	35.94
SW F-stat 1	68.07	68.07	68.07
SW F-stat 2	58.00	58.00	58.00
County FE	Y	Y	Y
Year FE	Y	Y	Y
Share Manufacturing	Y	Y	Y
Share Agriculture	Y	Y	Y
Active Coal Mines	Y	Y	Y

Replicated version

Figure 10: Table 4

- Table 5: different replicated output

	Dependent variable:			
	Any Union Present (1)	Number of Union Branches (2)	Union Density (Members / LF) (3)	Union Members per Branch (4)
<i>Panel A: Skilled Workers</i>				
Share of Immigrants	2.543** (1.162)	4.898*** (1.760)	0.753*** (0.237)	463.820*** (153.119)
Dep. var. mean	0.438	2.119	0.069	47.552
<i>Panel B: Unskilled Workers</i>				
Share of Immigrants	1.302 (1.057)	1.107 (1.113)	-0.001 (0.043)	202.708 (138.872)
Dep. var. mean	0.208	0.402	0.015	27.273
Observations	2,628	2,628	2,628	2,628
KP F-statistic	67.34	67.34	67.34	67.34

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	<code>l_comb_sk_pr</code>	<code>comb_sk_ihs</code>	<code>comb_sk_de</code>	<code>comb_sk_avgmemb</code>
Immigrant Share	2.622** (1.160)	4.983*** (1.735)	0.779*** (0.232)	467.635*** (151.378)
Observations	2,628	2,628	2,628	2,628
Outcome mean	0.440	2.147	0.070	47.948
Imm. Share mean	0.028	0.028	0.028	0.028
KP F-stat	67.36	67.36	67.36	67.36
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel A

	(1)	(2)	(3)	(4)
VARIABLES	<code>comb_unsk_pr</code>	<code>comb_unsk_ih</code>	<code>comb_unsk_dmb</code>	<code>comb_unsk_avgmemb</code>
Immigrant Share	1.355 (1.054)	1.147 (1.109)	0.010 (0.040)	213.984 (137.090)
Observations	2,628	2,628	2,628	2,628
Outcome mean	0.209	0.403	0.015	27.464
Imm. Share mean	0.028	0.028	0.028	0.028
KP F-stat	67.36	67.36	67.36	67.36
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel B

Figure 11: Table 5

- Table 6: different replicated output

Table 6: Heterogeneous Effects by Immigrant Labor Competition

	<i>Dependent variable:</i>			
	Any Union Present (1)	Number of Union Branches (2)	Union Density (Members / LF) (3)	Union Members per Branch (4)
<i>Panel A: Skilled Workers</i>				
Share of Immigrants [1]	2.228** (1.002)	3.974*** (1.336)	0.648*** (0.216)	365.007*** (127.993)
Share of Immigrants x Competition [2]	1.005** (0.465)	3.123** (1.211)	0.343** (0.137)	321.460*** (120.482)
KP F-statistic	46.40	46.40	46.40	46.40
LM F-statistic	38.44	38.44	38.44	38.44
SW F-statistic [1]	82.34	82.34	82.34	82.34
SW F-statistic [2]	45.65	45.65	45.65	45.65
<i>Panel B: Unskilled Workers</i>				
Share of Immigrants [1]	2.555* (1.304)	2.591* (1.375)	0.032 (0.048)	356.032* (185.777)
Share of Immigrants x Competition [2]	-0.769*** (0.296)	-0.910*** (0.346)	-0.020 (0.015)	-94.496** (40.949)
KP F-statistic	39.20	39.20	39.20	39.20
LM F-statistic	10.97	10.97	10.97	10.97
SW F-statistic [1]	80.32	80.32	80.32	80.32
SW F-statistic [2]	87.97	87.97	87.97	87.97
Observations	2,624	2,624	2,624	2,624

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	$l_comb_sk_prfl$	$comb_sk_lc$	$comb_sk_de$	$comb_sk_avgmemb$
Immigrant Share	2.336** (1.004)	4.151*** (1.320)	0.680*** (0.212)	360.315*** (126.905)
$sheu_co_euall_nat_mhs_s$	0.925** (0.470)	2.873** (1.183)	0.323** (0.133)	346.494*** (119.865)
Observations	2,624	2,624	2,624	2,624
Outcome mean	0.441	2.150	0.070	47.959
Imm. Share mean	0.028	0.028	0.028	0.028
KP F-stat	46.42	46.42	46.42	46.42
LM F-stat	38.42	38.42	38.42	38.42
SW F-stat 1	82.37	82.37	82.37	82.37
SW F-stat 2	45.66	45.66	45.66	45.66
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel A

	(1)	(2)	(3)	(4)
VARIABLES	comb_unsk_p	comb_unsk_	comb_unsk_dmb	comb_unsk_avgmemb
Immigrant Share	2.612** (1.300)	2.578* (1.371)	0.039 (0.046)	358.246* (183.474)
sheu_co_euall_nat_lsk_s	-0.771*** (0.294)	-0.878*** (0.339)	-0.018 (0.014)	-88.871** (40.424)
Observations	2,624	2,624	2,624	2,624
Outcome mean	0.209	0.404	0.015	27.505
Imm. Share mean	0.028	0.028	0.028	0.028
KP F-stat	39.20	39.20	39.20	39.20
LM F-stat	10.97	10.97	10.97	10.97
SW F-stat 1	80.33	80.33	80.33	80.33
SW F-stat 2	87.89	87.89	87.89	87.89
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel B

Figure 12: Table 6

- Table 7: different replicated output

Table 7: Heterogeneous Effects by Origin of Immigrants

	<i>Dependent variable:</i>			
	Any Union Present (1)	Number of Union Branches (2)	Union Density (Members / LF) (3)	Union Members per Branch (4)
Share of S/E European Immigrants	3.708* (2.197)	8.222** (4.031)	0.412** (0.209)	660.666** (305.790)
<i>Standardized coefficient</i>	[0.234]	[0.220]	[0.170]	[0.278]
Share of N/W European Immigrants	-0.209 (2.880)	-2.548 (4.589)	-0.058 (0.398)	10.681 (506.582)
<i>Standardized coefficient</i>	[-0.008]	[-0.042]	[-0.015]	[0.003]
Observations	2,628	2,628	2,628	2,628
KP F-statistic	10.42	10.42	10.42	10.42
LM F-statistic	12.35	12.35	12.35	12.35
SW F-statistic (S/E Europe)	22.78	22.78	22.78	22.78
SW F-statistic (N/W Europe)	39.68	39.68	39.68	39.68

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
sh_newsend_wa_10yr_m	3.687* (2.185) [0.232]	7.969** (3.919) [0.212]	0.431** (0.205) [0.176]	690.825** (306.595) [0.291]
sh_oldsend_wa_10yr_m	0.071 (2.876) [0.003]	-1.529 (4.517) [-0.025]	-0.055 (0.399) [-0.014]	-51.811 (508.658) [-0.013]
Observations	2,628	2,628	2,628	2,628
Outcome mean	0.444	2.589	0.037	48.577
Imm. Share S/E Europe mean	0.016	0.016	0.016	0.016
Imm. Share N/W Europe mean	0.012	0.012	0.012	0.012
KP F-stat	10.42	10.42	10.42	10.42
LM F-stat	12.35	12.35	12.35	12.35
SW F-stat 1	22.78	22.78	22.78	22.78
SW F-stat 2	39.70	39.70	39.70	39.70
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version

Figure 13: Table 7

- Table 8: different replicated output

Table 8: Heterogeneous Effects by Attitudes Towards Immigration

	<i>Dependent variable:</i>			
	Any Union Present (1)	Number of Union Branches (2)	Union Density (Members / LF) (3)	Union Members per Branch (4)
<i>Panel A: V = Vote share Know-Nothing party</i>				
Share of Immigrants [1]	1.631 (1.612)	-0.031 (2.390)	0.039 (0.183)	378.471 (306.528)
Share of Immigrants × High V [2]	1.182 (1.386)	5.477** (2.239)	0.203 (0.146)	162.836 (246.915)
<i>Sum of coefficients: [1] + [2]</i>	2.813	5.446*	0.242*	541.306**
Observations	2,103	2,103	2,103	2,103
KP F-statistic	23.63	23.63	23.63	23.63
LM F-statistic	18.46	18.46	18.46	18.46
SW F-statistic [1]	39.15	39.15	39.15	39.15
SW F-statistic [2]	86.90	86.90	86.90	86.90
<i>Panel B: V = Index of residential segregation</i>				
Share of Immigrants [1]	0.275 (1.327)	1.259 (2.139)	-0.055 (0.162)	-64.595 (188.162)
Share of Immigrants × High V [2]	2.769** (1.173)	4.603** (1.919)	0.403** (0.159)	643.350*** (175.594)
<i>Sum of coefficients: [1] + [2]</i>	3.044**	5.863***	0.348***	578.755***
Observations	2,565	2,565	2,565	2,565
KP F-statistic	31.71	31.71	31.71	31.71
LM F-statistic	18.34	18.34	18.34	18.34
SW F-statistic [1]	48.30	48.30	48.30	48.30
SW F-statistic [2]	74.83	74.83	74.83	74.83

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
Immigrant Share	1.818 (1.642)	0.387 (2.393)	0.039 (0.181)	366.304 (307.794)
sh_euro_knownot_t23	1.002 (1.407)	5.071** (2.233)	0.190 (0.145)	176.890 (247.940)
Observations	2,103	2,103	2,103	2,103
Outcome mean	0.457	2.656	0.039	51.989
Imm. Share mean	0.023	0.023	0.023	0.023
KP F-stat	23.64	23.64	23.64	23.64
LM F-stat	18.45	18.45	18.45	18.45
SW F-stat 1	39.17	39.17	39.17	39.17
SW F-stat 2	86.94	86.94	86.94	86.94
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y
[1] + [2]	2.820	5.458	0.229	543.194
s.e.	1.708	2.848	0.124	212.782

Replicated version, panel A

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
Immigrant Share	0.415 (1.356)	1.119 (1.977)	-0.019 (0.153)	-31.416 (178.665)
sh_euro_segr_en_t23	2.666** (1.200)	4.859*** (1.748)	0.377** (0.151)	609.552*** (167.177)
Observations	2,565	2,565	2,565	2,565
Outcome mean	0.448	2.624	0.037	49.253
Imm. Share mean	0.028	0.028	0.028	0.028
KP F-stat	31.73	31.73	31.73	31.73
LM F-stat	18.33	18.33	18.33	18.33
SW F-stat 1	48.32	48.32	48.32	48.32
SW F-stat 2	74.86	74.86	74.86	74.86
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y
[1] + [2]	3.081	5.978	0.358	578.136
s.e.	1.194	2.043	0.104	161.756

Replicated version, panel B

Figure 14: Table 8

- Table A3: different replicated output

Table A.3: Unionization and Immigration Flows

	<i>Dependent variable:</i> Share of Immigrants (x 100)			
	(1)	(2)	(3)	(4)
Any Union Present (t-10)	-0.660*** (0.252)			
Number of Union Branches (t-10)		-0.372** (0.161)		
Union Density (t-10)			-4.276* (2.418)	
Union Members per Branch (t-10)				-0.004** (0.002)
Observations	2,626	2,626	2,626	2,626
Dep. var. mean	2.788	2.788	2.788	2.788
Indep. var. mean	0.441	2.564	0.036	48.142

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	n_euro_wa_10yr_m	n_euro_wa_10yr_m	n_euro_wa_10yr_m	n_euro_wa_10yr_m
afl_comb_all_pres	-0.690*** (0.264)			
afl_comb_all_ihsloc		-0.381** (0.165)		
afl_comb_all_dens			-4.122* (2.391)	
afl_comb_all_avgmemb				-0.005** (0.002)
Observations	2,627	2,627	2,627	2,627
Outcome mean	2.788	2.788	2.788	2.788
Unionization mean	0.444	2.590	0.037	48.596
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version

Figure 15: Table A3

- Table A4: different replicated output

Table A.4: The Effect of Immigration on Organized Labor – Control Function

	<i>Dependent variable:</i>			
	Any Union Present (1)	Number of Union Branches (2)	Union Density (Members / LF) (3)	Union Members per Branch (4)
Share of Immigrants	2.626** (1.053)	5.245*** (1.820)	0.282*** (0.104)	481.007*** (147.736)
First-Stage Residuals of Share of Immigrants	-2.241** (1.128)	-3.544* (2.032)	-0.216* (0.110)	-347.835** (163.125)
Observations	2,628	2,628	2,628	2,628
Dep. var. mean	0.441	2.563	0.036	48.158
Indep. var. mean	0.028	0.028	0.028	0.028

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
Immigrant Share	2.687** (1.049)	5.344*** (1.794)	0.297*** (0.100)	485.556*** (146.540)
Share of Immigrants (first-stage residuals)	-2.320** (1.112)	-3.572* (1.993)	-0.222** (0.109)	-362.032** (162.434)
Nr. observations	2628	2628	2628	2628
Dep. var. mean	0.444	2.589	0.037	48.577
Indep. var. mean	0.028	0.028	0.028	0.028
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version

Figure 16: Table A4

- Table A5: different replicated output

Table A.5: Heterogeneous Effects by Immigrant Labor Competition

	<i>Dependent variable:</i>			
	Any Union Present (1)	Number of Union Branches (2)	Union Density (Members / LF) (3)	Union Members per Branch (4)
Share of Immigrants	3.297*** (1.234)	5.666*** (1.807)	0.306*** (0.116)	508.250*** (165.461)
Share of Immigrants × Skilled Comp.	0.677 (0.526)	3.356** (1.308)	0.155** (0.074)	301.723** (120.398)
Share of Immigrants × Unskilled Comp.	-0.540** (0.274)	-0.879* (0.495)	-0.044 (0.039)	-74.552* (43.503)
Observations	2,624	2,624	2,624	2,624
KP F-statistic	32.66	32.66	32.66	32.66
SW F-statistic (Share of Immigrants)	84.62	84.62	84.62	84.62
SW F-statistic (Sh. of Imm. × Sk. Comp.)	42.81	42.81	42.81	42.81
SW F-statistic (Sh. of Imm. × Unsk. Comp.)	50.93	50.93	50.93	50.93

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	<i>l_comb_all_prfl</i>	<i>l_comb_all_lc</i>	<i>l_comb_all_de</i>	<i>l_comb_all_avgmemb</i>
Immigrant Share	3.369*** (1.235)	5.835*** (1.790)	0.316*** (0.115)	511.046*** (164.625)
<i>sheu_co_euall_nat_mhsk_s</i>	0.642 (0.530)	3.089** (1.276)	0.148** (0.071)	323.136*** (120.413)
<i>sheu_co_euall_nat_lsk_s</i>	-0.539** (0.273)	-0.862* (0.479)	-0.040 (0.037)	-77.825* (43.900)
Observations	2,624	2,624	2,624	2,624
Outcome mean	0.444	2.592	0.037	48.589
Imm. Share mean	0.028	0.028	0.028	0.028
KP F-stat	32.67	32.67	32.67	32.67
SW F-stat 1	84.63	84.63	84.63	84.63
SW F-stat 2	42.83	42.83	42.83	42.83
SW F-stat 3	50.91	50.91	50.91	50.91
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version

Figure 17: Table A5

- Table A6: different replicated output

Table A.6: Heterogeneous Effects by Terciles of Attitudes Towards Immigration

	<i>Dependent variable:</i>			
	Any Union Present (1)	Number of Union Branches (2)	Union Density (Members / LF) (3)	Union Members per Branch (4)
<i>Panel A: V = Vote share Know-Nothing party</i>				
Share of Immigrants x Low V	1.535 (1.610)	-0.297 (2.460)	0.023 (0.185)	403.200 (313.316)
Share of Immigrants x Medium V	2.385 (1.775)	4.257 (3.631)	0.174 (0.126)	651.682** (285.172)
Share of Immigrants x High V	3.073 (2.043)	6.171* (3.175)	0.283* (0.146)	473.972** (209.739)
Observations	2,103	2,103	2,103	2,103
KP F-statistic	14.63	14.63	14.63	14.63
SW F-statistic (Low V)	37.92	37.92	37.92	37.92
SW F-statistic (Medium V)	40.74	40.74	40.74	40.74
SW F-statistic (High V)	42.35	42.35	42.35	42.35
<i>Panel B: V = Index of residential segregation</i>				
Share of Immigrants x Low V	0.289 (1.322)	1.533 (2.148)	-0.043 (0.160)	-57.646 (188.220)
Share of Immigrants x Medium V	3.208** (1.275)	9.169*** (2.613)	0.495*** (0.129)	662.753*** (199.877)
Share of Immigrants x High V	2.897** (1.416)	2.879 (1.873)	0.216** (0.104)	502.944*** (169.165)
Observations	2,565	2,565	2,565	2,565
KP F-statistic	21.09	21.09	21.09	21.09
SW F-statistic (Low V)	48.72	48.72	48.72	48.72
SW F-statistic (Medium V)	62.85	62.85	62.85	62.85
SW F-statistic (High V)	48.84	48.84	48.84	48.84

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
sh_euro_knownnot_t1	1.727 (1.642)	0.127 (2.460)	0.024 (0.183)	386.008 (314.429)
sh_euro_knownnot_t2	2.418 (1.773)	4.295 (3.573)	0.162 (0.124)	631.140** (283.596)
sh_euro_knownnot_t3	3.065 (2.029)	6.167** (3.131)	0.271* (0.144)	489.542** (212.552)
Observations	2,103	2,103	2,103	2,103
Outcome mean	0.457	2.656	0.039	51.989
Imm. Share mean	0.023	0.023	0.023	0.023
KP F-stat	14.64	14.64	14.64	14.64
SW F-stat 1	37.94	37.94	37.94	37.94
SW F-stat 2	40.76	40.76	40.76	40.76
SW F-stat 3	42.37	42.37	42.37	42.37
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel A

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
sh_euro_segr_en_t1	0.428 (1.351)	1.384 (1.986)	-0.007 (0.152)	-25.066 (178.952)
sh_euro_segr_en_t2	3.237** (1.276)	9.187*** (2.603)	0.500*** (0.128)	654.907*** (199.685)
sh_euro_segr_en_t3	2.940** (1.412)	3.081* (1.826)	0.231** (0.097)	508.847*** (168.230)
Observations	2,565	2,565	2,565	2,565
Outcome mean	0.448	2.624	0.037	49.253
Imm. Share mean	0.028	0.028	0.028	0.028
KP F-stat	21.10	21.10	21.10	21.10
SW F-stat 1	48.74	48.74	48.74	48.74
SW F-stat 2	62.87	62.87	62.87	62.87
SW F-stat 3	48.86	48.86	48.86	48.86
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel B

Figure 18: Table A6

- Table A7: different replicated output

Table A.7: Effect on the Composition of Union Leaders

	Dependent variable: Share of Union Leaders					
	U.S. (1)	N/W Europe (2)	S/E Europe (3)	U.S. (4)	N/W Europe (5)	S/E Europe (6)
	All counties			Always unionized counties		

Panel A: Origin country

Share of Immigrants	2.382** (1.064)	0.601 (0.398)	0.046 (0.134)	0.131 (0.277)	-0.142 (0.236)	0.068 (0.177)
Observations	2,628	2,628	2,628	579	579	579
Dep. var. mean	0.340	0.031	0.009	0.870	0.089	0.024
KP F-statistic	67.34	67.34	67.34	13.98	13.98	13.98

Panel B: Ancestry

Share of Immigrants	2.678** (1.159)	0.476 (0.362)	0.006 (0.398)	0.144 (0.389)
Dep. var. mean	2,625	2,625	579	579
Observations	0.342	0.038	0.881	0.101
KP F-statistic	67.27	67.27	13.98	13.98

Paper version

	(1)	(2)	(3)
VARIABLES	b_all_del_sh_o	b_all_del_sh_bb	b_all_del_sh_bpl_euse
Immigrant Share	2.446** (1.059)	0.607 (0.398)	0.038 (0.134)
Observations	2,628	2,628	2,628
Outcome mean	0.343	0.031	0.009
Imm. Share mean	0.028	0.028	0.028
KP F-stat	67.36	67.36	67.36
County FE	Y	Y	Y
Year FE	Y	Y	Y
Share Manufacturing	Y	Y	Y
Share Agriculture	Y	Y	Y
Active Coal Mines	Y	Y	Y

Replicated version, panel A

	(1)	(2)	(3)
VARIABLES	b_all_del_sh	b_all_del_sh	b_all_del_sh_bpl_euse
Immigrant Share	0.142 (0.278)	-0.137 (0.233)	0.016 (0.174)
Observations	582	582	582
Outcome mean	0.870	0.089	0.024
Imm. Share mean	0.046	0.046	0.046
KP F-stat	13.98	13.98	13.98
County FE	Y	Y	Y
Year FE	Y	Y	Y
Share Manufacturing	Y	Y	Y
Share Agriculture	Y	Y	Y
Active Coal Mines	Y	Y	Y

Replicated version, panel A (cont'd)

	(1)	(2)
VARIABLES	b_all_del_sh	b_all_del_sh_anc_euse
Immigrant Share	2.721** (1.144)	0.502 (0.371)
Observations	2,628	2,628
Outcome mean	0.337	0.043
Imm. Share mean	0.028	0.028
KP F-stat	67.36	67.36
County FE	Y	Y
Year FE	Y	Y
Share Manufacturing	Y	Y
Share Agriculture	Y	Y
Active Coal Mines	Y	Y

Replicated version, panel B

	(1)	(2)
VARIABLES	o_all_del_sh_ap	o_all_del_sh_anc_euse
Immigrant Share	0.080 (0.408)	0.052 (0.399)
Observations	582	582
Outcome mean	0.866	0.113
Imm. Share mean	0.046	0.046
KP F-stat	13.98	13.98
County FE	Y	Y
Year FE	Y	Y
Share Manufacturing	Y	Y
Share Agriculture	Y	Y
Active Coal Mines	Y	Y

Replicated version, panel B (cont'd)

Figure 19: Table A7

- Table A8: different replicated output

Table A.8: Heterogeneous Effects by Strength of Labor Movement in Country of Origin

	<i>Dependent variable:</i>			
	Any Union Present (1)	Number of Union Branches (2)	Union Density (Members / LF) (3)	Union Members per Branch (4)
Share of Immigrants from UK-Ireland	-4.457 (12.793)	-13.908 (23.535)	-0.872 (1.387)	-2,222.999 (2,057.525)
<i>Standardized coefficient</i>	[-0.046]	[-0.061]	[-0.059]	[-0.153]
Share of Immigrants from Other Countries	2.861* (1.507)	5.881** (2.600)	0.321** (0.139)	570.784*** (219.008)
<i>Standardized coefficient</i>	[0.231]	[0.202]	[0.169]	[0.308]
Observations	2,628	2,628	2,628	2,628
KP F-statistic	11.89	11.89	11.89	11.89
SW F-statistic (UK-Ireland)	26.27	26.27	26.27	26.27
SW F-statistic (Other Countries)	44.33	44.33	44.33	44.33

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
sh_unionh_wa_10yr_m	-5.461 (12.728) [-0.056]	-12.283 (23.199) [-0.054]	-0.958 (1.398) [-0.064]	-2,464.281 (2,080.045) [-0.170]
sh_unionl_wa_10yr_m	2.958** (1.500) [0.239]	5.929** (2.569) [0.203]	0.338** (0.136) [0.177]	583.496*** (219.523) [0.315]
Observations	2,628	2,628	2,628	2,628
Outcome mean	0.444	2.589	0.037	48.577
Imm. Share mean UK-Ireland	0.003	0.003	0.003	0.003
Imm. Share mean Other countries	0.025	0.025	0.025	0.025
KP F-stat	11.89	11.89	11.89	11.89
SW F-stat 1	26.28	26.28	26.28	26.28
SW F-stat 2	44.35	44.35	44.35	44.35
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version

Figure 20: Table A8

- Table A9: different replicated output

Table A.9: Heterogeneous Effects by Support for Socialist Parties in Country of Origin

	Dependent variable:			
	Any Union Present (1)	Number of Union Branches (2)	Union Density (Members / LF) (3)	Union Members per Branch (4)
<i>Panel A: High Vote Share = Above 20% during 1890-1919</i>				
Share of Immigrants from Countries w/ High Socialist Vote Share	1.013 (2.559) [0.044]	-1.213 (4.339) [-0.023]	0.045 (0.271) [0.013]	-194.233 (392.421) [-0.057]
Standardized coefficient				
Share of Immigrants from Countries w/ Low Socialist Vote Share	3.582 (2.675) [0.190]	9.078** (4.117) [0.204]	0.424** (0.213) [0.147]	881.742*** (326.762) [0.312]
Standardized coefficient				
KP F-statistic	14.42	14.42	14.42	14.42
SW F-statistic (High Socialist Vote Share)	27.76	27.76	27.76	27.76
SW F-statistic (Low Socialist Vote Share)	38.00	38.00	38.00	38.00
<i>Panel B: High Vote Share = Above 10% during 1890-1919</i>				
Share of Immigrants from Countries w/ High Socialist Vote Share	-1.825 (2.020) [-0.131]	-5.299 (3.571) [-0.161]	-0.156 (0.163) [-0.073]	-298.928 (288.394) [-0.143]
Standardized coefficient				
Share of Immigrants from Countries w/ Low Socialist Vote Share	25.893* (14.280) [0.639]	60.369*** (21.532) [0.632]	2.577*** (0.842) [0.416]	4,558.319*** (1,544.896) [0.751]
Standardized coefficient				
KP F-statistic	3.93	3.93	3.93	3.93
SW F-statistic (High Socialist Vote Share)	12.11	12.11	12.11	12.11
SW F-statistic (Low Socialist Vote Share)	7.98	7.98	7.98	7.98
Observations	2,628	2,628	2,628	2,628

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
sh_soc20_wa_10yr_m	1.187 (2.559) [0.052]	-0.782 (4.296) [-0.014]	0.075 (0.266) [0.021]	-232.279 (386.477) [-0.068]
sh_nonsoc20_wa_10yr_m	3.578 (2.672) [0.189]	8.979** (4.064) [0.201]	0.428** (0.209) [0.147]	911.571*** (327.625) [0.322]
Observations	2,628	2,628	2,628	2,628
Outcome mean	0.444	2.589	0.037	48.577
Imm. Share mean Socialist	0.011	0.011	0.011	0.011
Imm. Share mean Non-socialist	0.017	0.017	0.017	0.017
KP F-stat	14.43	14.43	14.43	14.43
SW F-stat 1	27.77	27.77	27.77	27.77
SW F-stat 2	38.01	38.01	38.01	38.01
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel A

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
sh_soc10_wa_10yr_m	-1.663 (2.003) [-0.119]	-4.808 (3.471) [-0.145]	-0.123 (0.160) [-0.057]	-332.107 (290.696) [-0.159]
sh_nonsoc10_wa_10yr_m	25.432* (14.201) [0.628]	58.413*** (21.143) [0.610]	2.490*** (0.835) [0.398]	4,760.104*** (1,548.614) [0.784]
Observations	2,628	2,628	2,628	2,628
Outcome mean	0.444	2.589	0.037	48.577
Imm. Share mean Socialist	0.021	0.021	0.021	0.021
Imm. Share mean Non-socialist	0.007	0.007	0.007	0.007
KP F-stat	3.93	3.93	3.93	3.93
SW F-stat 1	12.11	12.11	12.11	12.11
SW F-stat 2	7.98	7.98	7.98	7.98
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel B

Figure 21: Table A9

- Table A10: different replicated output

Table A.10: Heterogeneous Effects by the Number of Other Unions' Branches

	<i>Dependent variable:</i>			
	Any Union Present (1)	Number of Union Branches (2)	Union Density (Members / LF) (3)	Union Members per Branch (4)
<i>Panel A: Heterogeneity by Knights of Labor (KoL) Branches</i>				
Share of Immigrants [1]	2.712** (1.233)	5.141** (2.026)	0.266** (0.104)	474.221*** (159.633)
Share of Immigrants × KoL Branches [2]	-0.329 (0.381)	0.481 (1.086)	0.064 (0.058)	24.687 (80.178)
Observations	2,616	2,616	2,616	2,616
KP F-statistic	33.21	33.21	33.21	33.21
SW F-statistic [1]	65.84	65.84	65.84	65.84
SW F-statistic [2]	68.78	68.78	68.78	68.78
<i>Panel B: Heterogeneity by Industrial Workers of the World (IWW) Branches</i>				
Share of Immigrants [1]	2.978** (1.258)	4.314** (1.921)	0.290** (0.114)	501.300*** (169.781)
Share of Immigrants × IWW Branches [2]	-0.403* (0.208)	1.063 (0.917)	-0.009 (0.031)	-23.157 (44.224)
Observations	2,628	2,628	2,628	2,628
KP F-statistic	32.27	32.27	32.27	32.27
SW F-statistic [1]	64.05	64.05	64.05	64.05
SW F-statistic [2]	65.37	65.37	65.37	65.37

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
Immigrant Share	2.769** (1.231)	5.233*** (1.998)	0.283*** (0.101)	479.804*** (157.784)
sheuro_KOL	-0.310 (0.382)	0.504 (1.068)	0.054 (0.056)	20.505 (80.661)
Observations	2,616	2,616	2,616	2,616
Outcome mean	0.444	2.586	0.037	48.627
Share Immigrants	0.028	0.028	0.028	0.028
KP F-stat	33.23	33.23	33.23	33.23
SW F-stat 1	65.86	65.86	65.86	65.86
SW F-stat 2	68.81	68.81	68.81	68.81
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel A

	(1)	(2)	(3)	(4)
VARIABLES	_comb_all_pr	_comb_all_ihs	_comb_all_de	_comb_all_avgmemb
Immigrant Share	3.049** (1.257)	4.444** (1.891)	0.308*** (0.110)	501.018*** (168.193)
sheuro_IWW	-0.411* (0.210)	1.021 (0.893)	-0.012 (0.030)	-17.562 (44.600)
Observations	2,628	2,628	2,628	2,628
Outcome mean	0.444	2.589	0.037	48.577
Share Immigrants	0.028	0.028	0.028	0.028
KP F-stat	32.25	32.25	32.25	32.25
SW F-stat 1	64.01	64.01	64.01	64.01
SW F-stat 2	65.34	65.34	65.34	65.34
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version, panel B

Figure 22: Table A10

- Table A11: different replicated output

Table A.11: Effect on Local Economic Outcomes

	<i>Dependent variable:</i>			
	Labor Force Part. Rate (1)	Mfg. Output (per Worker) (2)	Mfg. Output (Share of U.S. Total) (3)	Share of LF in Skilled Occupations (4)
Share of Immigrants	-0.003 (0.098)	-0.866 (1.094)	-0.002 (0.015)	0.104 (0.110)
<i>Standardized coefficient</i>	[-0.003]	[-0.072]	[-0.031]	[0.048]
Observations	2,628	2,595	2,595	2,628
Outcome mean	0.909	3163.887	0.001	0.201
Imm. Share mean	0.028	0.028	0.028	0.028
KP F-statistic	67.34	66.68	66.68	67.34

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	lfpartrate_m	og_mfgout	pvshare_mfgout	elf_midhighsk2_m
Immigrant Share	-0.003 (0.098) [-0.003]	-0.866 (1.094) [-0.072]	-0.002 (0.015) [-0.031]	0.104 (0.110) [0.048]
Observations	2,628	2,595	2,595	2,628
Outcome mean	0.909	3163.887	0.001	0.201
Imm. Share mean	0.028	0.028	0.028	0.028
KP F-stat	67.36	66.71	66.71	67.36
Share Manufacturing	Y	Y	Y	Y
Share Agriculture	Y	Y	Y	Y
Active Coal Mines	Y	Y	Y	Y

Replicated version

Figure 23: Table A11

- Table B1: different replicated output

Table B.1: First Stage of the Alternative Instrumental Variable Estimation

	<i>Dependent variable:</i> Share of Immigrants			
	(1)	(2)	(3)	(4)
Predicted Share of Immigrants	0.211*** (0.042)	0.183*** (0.039)	0.166*** (0.035)	0.166*** (0.034)
Observations	2,628	2,628	2,628	2,628
Dep. var. mean	0.028	0.028	0.028	0.028
Indep. var. mean	0.094	0.094	0.094	0.094
KP F-statistic	25.30	22.09	23.12	23.60
<i>Controls:</i>				
Share of Population in Manufacturing	No	Yes	Yes	Yes
Share of Population in Agriculture	No	No	Yes	Yes
Presence of Coal Mines	No	No	No	Yes

Paper version

	(1)	(2)	(3)	(4)
VARIABLES	Immigrant Share	Immigrant Share	Immigrant Share	Immigrant Share
pr_temp_sh_euro_10yr_m	0.211*** (0.042)	0.183*** (0.039)	0.166*** (0.035)	0.166*** (0.034)
Observations	2,628	2,628	2,628	2,628
Outcome mean	0.028	0.028	0.028	0.028
Imm. Share mean	0.094	0.094	0.094	0.094
KP F-stat	25.31	22.10	23.13	23.61
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Share Manufacturing	N	Y	Y	Y
Share Agriculture	N	N	Y	Y
Active Coal Mines	N	N	N	Y

Replicated version

Figure 24: Table B1

- Table G1: different replicated output

Table G.1: Summary of Rotemberg Weights

Panel A: Negative and positive weights

	Sum	Mean	Share
Negative	-0.062	-0.015	0.055
Positive	1.062	0.082	0.945

Panel B: Correlations

	$\hat{\alpha}_j$	g_j	$\hat{\beta}_j$	$\hat{F}_j$	Var(z_j)
$\hat{\alpha}_j$	1				
g_j	0.918	1			
$\hat{\beta}_j$	0.126	-0.070	1		
$\hat{F}_j$	0.115	0.124	0.055	1	
Var(z_j)	-0.222	-0.391	0.475	-0.383	1

Panel C: Top five Rotemberg weight countries of origin

	$\hat{\alpha}_j$	g_j	$\hat{\beta}_j$	95% CI
Austria-Hungary	0.267	7.55e+05	0.305	(-1.100,1.700)
Italy	0.258	7.36e+05	-1.135	(-4.500,-1.000)
Russia	0.227	6.62e+05	4.447	(-0.500,2.100)
Greece-Portugal-Spain	0.100	1.06e+05	13.431	(-3.200,3.100)
Sweden	0.063	1.20e+05	0.985	(-0.700,2.200)

Panel D: Estimates of β_j for positive and negative weights

	α -weighted sum	Share of overall β	Mean
Negative	0.192	0.072	2.021
Positive	2.461	0.928	1.648

Paper version

Panel A: Negative and positive weights

	Sum	Mean	Share
Negative	-0.062	-0.015	0.055
Positive	1.062	0.082	0.945

Panel B: Correlations

	α_k	g_k	β_k	F_k	$\text{Var}(z_k)$
α_k	1				
g_k	0.918	1			
β_k	0.119	-0.079	1		
F_k	0.115	0.124	0.053	1	
$\text{Var}(z_k)$	-0.222	-0.391	0.496	-0.383	1

Panel C: Top 5 Rotemberg weight countries of origin

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95 % CI
Austria-Hungary	0.267	7.55e+05	0.391	(-2.200,3.300)
Italy	0.258	7.36e+05	-1.072	(-4.800,1.100)
Russia	0.227	6.62e+05	4.511	(1.800,15.200)
Greece-Portugal-Spain	0.100	1.06e+05	13.427	(0.500,56.200)
Sweden	0.063	1.20e+05	1.189	(-2.600,4.600)

Panel D: Estimates of β_k for positive and negative weights

	α -weighted Sum	Share of overall β	Mean
Negative	0.168	0.062	2.424
Positive	2.547	0.938	1.683

Replicated version

Figure 25: Table G1

9.4 In-Text Numbers

In-text numbers	Program	Line Number	Reproduced?
p.3 & 39: "in the absence of immigration, union density 1900-1920 would have been ~22% lower" (also ~31% for skilled)	2c_results_main.do	195–197	Yes

- ☐ In-text numbers not verified.
- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code

9.5 Other Issues

- **README.md** did not enlist raw data but rather intermediate datasets and their sources. Even though they are not shipped in the package, it is important to cite them as if they were included.

10 Classification

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☒ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☒ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements

10.2 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and*

documented) consent of individuals. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary:

- Data files with PII indicators: 21
- Variables flagged in data: 114
- Code files with PII references: 39
- PII references in code: 2663

10.2.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	Crosswalk_1880_1930.dta	1	lat
Data	Crosswalk_1890_1930.dta	1	lat
Data	Crosswalk_1900_1930.dta	1	lat
Data	Crosswalk_1910_1930.dta	1	lat
Data	Crosswalk_1920_1930.dta	1	lat
Data	IWW_unions_county1930.csv	1	loc
Data	KoL_unions_county1930.csv	1	loc
Data	Railroad_Ataack_1930countyboundaries.csv	1	name
Data	Weather_Data.dta	8	son, lat, lon, country, name
Data	agri_census_1890.csv	1	name
Data	coalmines_1890.csv	2	name, loc
Data	electionresults_1856_county.csv	1	name
Data	electionresults_1886-1924_county.csv	1	name
Data	mfg_census.csv	1	name
Data	mines_census_1890.csv	2	name, loc
Data	names_ancestry.dta	26	name
Data	names_origin.dta	27	name

File Type	File	Variables/References	PII Categories
Data	unions_combined_county1930.csv	34	loc
Data	willcox_immigration_bycountry.csv	1	country
Data	xwalk_occ1950_occnames.csv	1	name

See [Appendix](#) for detailed listing of all flagged instances.